Started on	Sunday, 24 July 2022, 4:37 PM
State	Finished
Completed on	Sunday, 24 July 2022, 4:38 PM
Time taken	53 secs
Grade	**0.00** out of 45.00 (**0**%)

Question **1**

Not answered

Marked out of 8.00

IV Haloperidol
Venlafaxine
Citalopram
Aripiprazole
Mirtazapine
Risperidone
Sertindole
Quetiapine

QTc prolongation
Mr Barnes is a 78-year-old man with a diagnosis of schizophrenia and moderate depression. He has been trialled on a number of medications, some of them with only limited success. You are reviewing his notes and deciding on the next possible pharmacotherapeutic options. You are aware he has a family history of cardiac dysfunction, with ischaemic heart disease on the maternal side, and cardiac arrhythmia on the paternal side. Choose from the list of medications given, that has:

1. No effect on the QTc at therapeutic doses (choose 2 options)

2. Low effect on the QTc at therapeutic doses (choose 2 options)

3. Moderate effect on the QTc at therapeutic doses (choose 2 options)

4. High effect on the QTc interval at therapeutic doses (choose 2 options)

1) **No effect** on the QTc at therapeutic doses- Drugs include **Aripiprazole** and **Mirtazapine**

2) **Low effect** on the QTc at therapeutic doses- Drugs include **Venlafaxine** and **Risperidone**.

3) **Moderate effect** on the QTc at therapeutic doses- Drugs include **Citalopram** and **Quetiapine**.

4) **High effect** on the QTc at therapeutic doses- Drugs include **Haloperidol**, **Sertindole** and **any intravenous antipsychotic.**

(Ref: The Maudsley Prescribing guidelines- 11th edition- page 117)

Question **2**

Not answered

Marked out of 1.00

Flag question

Veronica has been going through a period of change in her life, and she's feeling particularly stressed. She is finding it especially difficult to get to sleep. When you take a careful history, you discover she's been drinking three cups of coffee before going to bed. Which is the main neuro-receptor implicated in Veronica's struggle to get to sleep after drinking a lot of coffee?

Select one:

- Dopamine receptor
- Adenosine receptor
- Serotonin receptor
- GABA receptor
- Opioid receptor

Your answer is incorrect.

Coffee's stimulant action is largely due to its action on adenosine receptors. When adenosine binds to its receptors, neural activity slows down and this is associated with feeling sleepy. Caffeine is an adenosine receptor antagonist and has the effect of speeding up neural activity and making us feel more alert.

The correct answer is: Adenosine receptor

Question **3**

Not answered

Marked out of 1.00

Flag question

Shaun is an 11-year-old child who has recently been diagnosed with epilepsy. He has been prescribed a medication to reduce the incidence of his seizures, but his parents are reporting that he has had some angry outbursts, behaving in a destructive manner, smashing some of his toys and frightening his siblings. Which of the following anti-epileptics is most likely to be associated with behavioural disturbances?

Select one:

- Gabapentin
- Pregabalin
- Lamotrigine
- Vigabatrin
- Valproate

Your answer is incorrect.

Of the medications listed, Vigabatrin is most associated with behavioural disturbances, particularly in children; a French study found behavioural disturbances in children taking Vigabatrin to be as high as 26% (Ref: Epilepsia Vol 47 Suppl 2, 2006).

The correct answer is: Vigabatrin

Question **4**

Not answered

Carol has been taking Paroxetine 40mg every day for the past two months in the hope that it may alleviate her feelings of depression. She thinks her mood has improved somewhat, and now she wishes to stop the antidepressant. She doesn't consult her GP, but she stops her

Marked out of 1.00

Flag question

medication abruptly. Over the next few days she feels unwell: tired, with aching muscles and joints, headaches and nauseated. The feeling pass after about a week. She reads up on why this might have happened and she thinks it might be to do with SSRI discontinuation syndrome. What is a proposed mechanism for SSRI Discontinuation Syndrome?

Select one:

- There is no physiological change, but it is all explained in psychological terms
- Temporary deficiency in the brain of one or more essential neurotransmitters associated with mood
- Release of inflammatory proteins producing flu-like symptoms
- Long-standing changes to brain structure which reduce the brain's capacity to produce neurotransmitters associated with mood
- Temporary excess in the brain of one or more essential neurotransmitters associated with mood

Your answer is incorrect.

Even though SSRI Discontinuation Syndrome is not completely understood, this answer is the most likely according to current research in the area. Neurotransmitters implicated include serotonin, dopamine and acetylcholine (cholinergic rebound). (Ref: Am Fam Physician 2006 1;74(3))

The correct answer is: Temporary deficiency in the brain of one or more essential neurotransmitters associated with mood

Question **5**

Not answered

Marked out of 1.00

Flag question

Paul has been taking Risperidone as recommended by her psychiatrist, following an admission to hospital during which she was experiencing auditory hallucinations. She has now noticed that her periods have stopped, and there is milk leaking from her breasts. Her blood investigations reveal a Prolactin level of 960 MiU/L. What dopamine receptor is most implicated in raised prolactin?

Select one:

- D1
- D9
- D5
- D3
- D2

Your answer is incorrect.

High affinity to D2, and slow dissociation from the receptor once bound, is associated with high prolactin levels. Dopamine itself has a central role in the regulation of prolactin secretion, a mechanism that is inhibited by several antipsychotics.

The correct answer is: D2

Question **6**

Not answered

Marked out of 1.00

Flag question

Andrew is a 32-year-old man. He presents with some emotional and behavioural disturbances of few months' duration that are out of character for him. His partner reports that Andrew has also been forgetful, and that his gait is awkward, with episodes of loss of balance. When Andrew sits down, his legs appear restless, and there are occasional non goal-directed arm movements that he does not seem to be able to control. What part of the history is most important to confirm?

Select one:

- Sexual history
- Social history
- Educational history
- Family history
- Forensic history

Your answer is incorrect.

The presentation is suggestive of a neurodegenerative illness such as Huntington's, which is genetically inherited, with a heritability of 50%. A family history of similar problems would be a very important component of the history to clarify.

The correct answer is: Family history

Question **7**

Not answered

Marked out of 1.00

Flag question

Meliah is a 19-year-old woman who presents with features of an anxiety disorder. You take a family history and find out that her mother presented with similar problems when she was in her late-20s, and her maternal grandmother when she was in her 40s. Apparently her great-grandmother also had it, beginning in her fifth decade. Genetic anticipation is associated with which anxiety disorder?

Select one:

- PTSD
- OCD
- Panic disorder
- Spider phobia
- Trichotillomania

Your answer is incorrect.

Based on family studies, it has been hypothesised that OCD and Obsessive-compulsive spectrum disorders showed significantly earlier age of onset than the parental generation (Ref: Psychiatric Research Vol 111, Issue 1, Aug 2002)

The correct answer is: OCD

Question **8**

Not answered

Marked out of

Janet's mother was diagnosed with Wilson's Disease and is concerned that she too will develop the condition. What inheritance pattern does Wilsons Disease follow?

1.00

Flag question

Select one:

- X-linked recessive
- Mitochondrial inheritance
- Autosomal dominant
- Autosomal recessive
- X-linked dominant

Your answer is incorrect.

Wilson's disease is an inherited condition (autosomal recessive), which results in accumulation of copper affecting certain organs such as the liver, brain and eyes. Characteristic Kayser-Fleischer rings are seen in 95% of patients who exhibit neurological signs of the disease.

The correct answer is: Autosomal recessive

Question **9**

Not answered

Marked out of 1.00

Flag question

Autistic spectrum disorder is associated with mutations in which of the following genes?

Select one:

- 22 q 11
- SHANK 3 gene
- HPRT gene
- 15 q 11-13
- FMR1 gene

Your answer is incorrect.

Mutations in the SHANK3 gene (located on chromosome 22) are associated with autistic spectrum disorder. Lesch-Nyhan Syndrome - mutations in HPRT1 gene. Fragile X - mutations in FMR1 gene. Di George Syndrome - 22q11.2 deletion. Prader-Willi Syndrome - deletions along region 15q11-13

The correct answer is: SHANK 3 gene

Question **10**

Not answered

Marked out of 1.00

Flag question

Which of the following is a X-linked recessive disorder?

Select one:

- Hunter syndrome
- Hurler syndrome
- Rett syndrome
- Kleinfelter's syndrome
- Huntington's disease

Your answer is incorrect.

Hunter syndrome is a mucopolysaccharidosis type II (MPS II) that is inherited (x-linked recessive). Symptoms, which appear after the infant reaches a year old include recurrent ear infections, runny noses and abdominal hernias. Facial features include coarse features, flattened bridge of nose and enlarged tongue.

The correct answer is: Hunter syndrome

Question **11**

Not answered

Marked out of 1.00

Flag question

Functional MRI scanning of a patient experiencing auditory hallucinations may show?

Select one:

- Increased activity in the thalamus
- Increased activity in frontal lobe
- Increased activity in the temporal cortex
- Reduced activity in hippocampus
- No changes

Your answer is incorrect.

Functional MRI scanning demonstrates increased cerebral blood flow in the right superior temporal gyrus and caudate and a decrease in the bilateral occipital and left parietal cortices. There is consistent cerebral blood flow decrease in people with schizophrenia in the bilateral occipital regions, the left lateral prefrontal and insular cortices, and the right anterior cingulate cortex. They also show increased blood flow in the bilateral lateral temporal regions and putamen, the left middle cingulate cortex and the right thalamus.

The correct answer is: Increased activity in the temporal cortex

Question **12**

Not answered

Marked out of 1.00

Flag question

Mr Keith Williams is a 40-year-old man with a history of alcohol dependence who is brought to the hospital by his family. They are concerned as he has been having difficulty walking and cannot remember any events over the last several days. Which of the following could be true for Mr Williams?

Select one:

- Increased ALT and AST levels
- Increased acetyldehyde dehydrogenase
- Decreased Gamma GT levels
- Decreased mamillary body transketolase activity
- Decreased aldehyde dehydrogenase

Your answer is incorrect.

The diagnosis is Wenicke-Korsakoff syndrome, which is associated with reduced activity of transketolase in the mammillary bodies. Transketolase activity is decreased in deficiency of thiamine, such as in Wernicke-Korsakoff syndrome. There is an indication that thiamine deficiency leads to Wernicke-Korsakoff syndrome only in those whose transketolase has a

reduced affinity for thiamine. In this way, the activity of transketolase is greatly hindered, and, as a consequence, the entire pentose phosphate pathway is inhibited.

The correct answer is: Decreased mamillary body transketolase activity

Question **13**

Not answered

Marked out of 1.00

Flag question

Which of the following functions is in the non-dominant lobe?

Select one:

- Spatial orientation
- Language
- Maths
- Reading
- Prosody

Your answer is incorrect.

Functions of the non-dominant lobes include: Parietal: spatial awareness, some motor control (e.g. constructional apraxia if damaged), perception of body (e.g. if damaged then anosognosia) Temporal: non-verbal memory, musical skills

The correct answer is: Spatial orientation

Question **14**

Not answered

Marked out of 1.00

Flag question

What area of the brain is blood flow increased in depression?

Select one:

- Calcarine sulcus
- Posterior occipital lobe
- Hippocampal sulcus
- Fusiform gyrus
- Anterior cingulate cortex

Your answer is incorrect.

The anterior cingulate cortex (ACC) has been implicated in the modulation of emotional behaviour. Functional neuroimaging studies have shown increased cerebral blood flow to this region in unmedicated patients with depression, as well as an increase in this area in healthy patients in times of anxiety and sadness.

The correct answer is: Anterior cingulate cortex

Question **15**

Not answered

Marked out of 1.00

Antonio has had poorly controlled diabetes since he was 5 years old. Now in his early 20s, he takes Captopril in order to control his hypertension, which has resulted from renal dysfunction. He is admitted to a psychiatric ward in a state of disinhibition, and appears elated, grandiose and psychotic. You are considering starting him on a medication to stabilise his mood. Which medication should be avoided in particular if we take into

Flag question

consideration his physical health problems and current medication?

Select one:

- Lamotrigine
- Soldium Valproate
- Lithium
- Risperidone
- Olanzapine

Your answer is incorrect.

Lithium is renally excreted and must be used with extreme caution in patients with impaired renal function, and in conjunction with medications such as ACE inhibitors, which may affect renal function. In such cases there's the risk that Lithium is not excreted at an adequate rate, leading to toxicity.

The correct answer is: Lithium

Question **16**

Not answered

Marked out of 1.00

Flag question

Mrs Myrtle has recently been discharged from a two-month admission to hospital. She was diagnosed with schizophrenia ten years ago and her concordance with oral medication has been poor. On this last admission she was started on a depot medication. At a follow-up meeting she is accompanied by her husband who reports that since this last admission she sometimes kicks him in bed. She also doesn't seem to be able to sit still in a chair. What is the best fitting description?

Select one:

- Parkinsonism
- Akathisia
- Tardive dyskinesia
- REM sleep disorder
- Aggressive disorder

Your answer is incorrect.

Antipsychotics can precipitate akathisia, a movement disorder characterised by restlessness, fidgeting and a difficulty in keeping still.

The correct answer is: Akathisia

Question **17**

Not answered

Marked out of 1.00

Flag question

Mr Farah was admitted to Psychiatric Intensive Unit in an acutely psychotic state, believing the MI5 to be persecuting him, and resulting in him lashing out at strangers in public places, causing physical injury. He is restrained and given an injection of an antipsychotic medication. Two hours later, when he is re-examined, he's found to be pyrexic, hypertensive, confused, with muscle rigidity. What is the most likely cause of this presentation?

Select one:

- Heamorrhagic stroke
- HIV Encephalitis
- Serotonin Syndrome
- Anticholinergic syndrome
- Neuroleptic Malignant Syndrome

Your answer is incorrect.

NMS is a rare but life-threatening reaction to neuroleptic medications characterised by muscle rigidity, fever, autonomic dysregulation, altered mental state.

The correct answer is: Neuroleptic Malignant Syndrome

Question **18**

Not answered

Marked out of 1.00

Flag question

Mr Samuel Hicks takes a regular antipsychotic medication and has monthly blood monitoring for this. He presents acutely with fever and chest pain. What ECG changes would you see?

Select one:

- Atrial bigeminy
- Prolongation of PR segment
- Tenting of T waves
- Evidence of deep Q waves
- Saddling of the ST segment

Your answer is incorrect.

Mr Hicks will be taking Clozapine as he is having regular blood tests for an antipsychotic prescription. This can cause myocarditis as a rare, but serious, side effect. This is characterised by saddled elevation of the ST segment, diffuse T wave inversion and other non-specific ST changes.

The correct answer is: Saddling of the ST segment

Question **19**

Not answered

Marked out of 1.00

Flag question

What is a risk factor for antidepressant-induced SIADH (syndrome of inappropriate ADH secretion)?

Select one:

- High initial sodium level
- Tuberculosis
- Cold weather
- Low body weight
- Male sex

Your answer is incorrect.

Risk factors for antidepressant-induced SIADH include: Old age, low BMI, concomitant use of diuretics and female gender.

The correct answer is: Low body weight

Question **20**

Not answered

Marked out of 1.00

Flag question

Rosie is a 25-year-old woman brought into A&E by her boyfriend who came home to find her very confused. He notes that she is on some medication but doesn't know what it is. Her vital signs include pyrexia and a labile blood pressure, and on examination clonus in ankles. The most likely diagnosis is:

Select one:

- Serotonin syndrome
- Meningitis
- Malignant hyperthermia
- Encephalitis
- Neuroleptic malignant syndrome

Your answer is incorrect.

Serotonin syndrome is the excess of serotonin resulting in a constellation of symptoms including confusion, agitation, lability in blood pressure and temperature, tremor and clonus. To differentiate between neuroleptic malignant syndrome and serotonin syndrome - onset for serotonin syndrome is much shorter (usually < 12hours), it is associated with hyperreflexia and clonus (NMS is associated with hyporeflexia), dilated pupils and also hyperactive bowel sounds.

(Ref: http://www.medsafe.govt.nz/profs/PUArticles/Dec2012Neuroleptic.htm)

The correct answer is: Serotonin syndrome

Question **21**

Not answered

Marked out of 1.00

Flag question

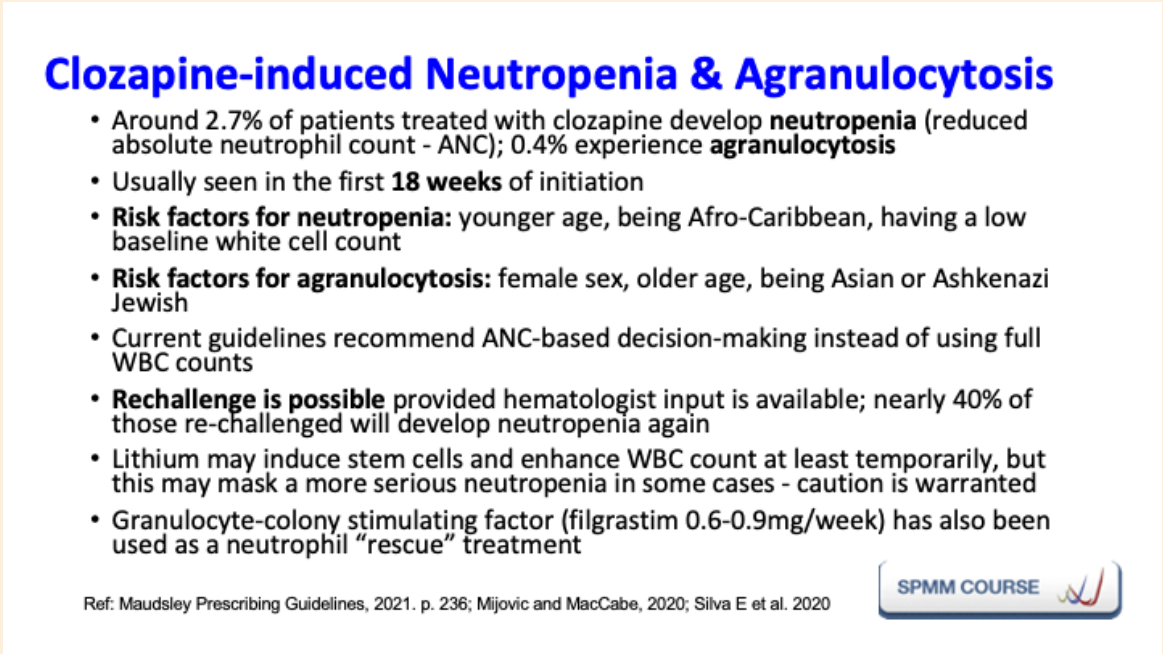
What is a risk factor for developing agranulocytosis with clozapine use?

Select one:

- Female sex
- Younger age
- Caucasian
- Low baseline white cell count
- Long term clozapine use

Your answer is incorrect.

Explanation:



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 236.

Mijovic A, MacCabe JH. Clozapine-induced agranulocytosis. Annals of Hematology. 2020 Aug 20:1-6.

Silva E et al. Clozapine rechallenge and initiation despite neutropenia-a practical, step-by-step guide. BMC psychiatry. 2020 Dec;20:1-9.

The correct answer is: Female sex

Question **22**

Not answered

Marked out of 1.00

Flag question

Ben Staines is a 42-year-old man with a diagnosis of schizophrenia, who has been stable on Clozapine for the past year. He presents to clinic reporting low mood for the past three weeks, with loss of energy, reduced interest in life. You consider starting him on an antidepressant. Which is best to avoid, given that he is on Clozapine?

Select one:

- Fluoxetine
- Selegiline
- Amitriptyline
- Sertraline
- Escitalopram

Your answer is incorrect.

Explanation:

Fluoxetine is a potent inhibitor of CYP2D6 and can raise clozapine levels, resulting in increased seizure risk with clozapine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 308-309.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

The correct answer is: Fluoxetine

Question **23**

Not answered

Marked out of 1.00

Flag question

In relation to weight gain with antipsychotics, which of the following statements is true:

Select one:

- It is more significant in children
- It is insignificant
- Occurs more with first generation antipsychotics
- Occurs less with second generation antipsychotics
- It is more significant in adults

Your answer is incorrect.

Explanation:

An increased susceptibility to antipsychotic-induced weight gain has been repeatedly documented in children, particularly rapid weight gain. Furthermore, antipsychotic medications touted to be weight-neutral in adults can cause substantial weight gain in children.

Ref: Libowitz MR, Nurmi EL. The burden of antipsychotic-induced weight gain and metabolic syndrome in children. Frontiers in Psychiatry. 2021 Mar 12;12:275.

The correct answer is: It is more significant in children

Question **24**

Not answered

Marked out of 1.00

Flag question

Sean is a 19 year old, previously healthy man, who was involved in a motorbike accident resulting in a head injury. He is generally recovering well. Regarding his cognitive capacities, he is able to read, write and speak spontaneously. He is not able, however, to comprehend speech. What is this describing?

Select one:

- Auditory Verbal Agnosia
- Prosopagnosia
- Anomia
- Apperceptive agnosia
- Associative agnosia

Your answer is incorrect.

Auditory Verbal Agnosia is also known as pure word deafness, this is the inability to comprehend speech, and includes not being able to repeat words and write from dictation. Other skills such as reading, writing, and spontaneous speech are preserved.

The correct answer is: Auditory Verbal Agnosia

Question **25**

You have a 76-year-old patient, Dennis Smythe, who has been experiencing some cognitive

Not answered

Marked out of 1.00

Flag question

impairment and personality changes and you wish to request a test that does not expose him to radiation, and that can map neuronal activity through the use of blood-oxygen-level dependent contrast. Which test do you choose?

Select one:

- F-MRI
- EEG
- CT of the head
- X-ray of the head
- Transcranial ultrasound

Your answer is incorrect.

f-MRIs have the advantage of not exposing patients to ionising radiation; it measures brain activity by detecting changes associated with blood flow, which gives an indication of neuronal activation; useful for investigating conditions such as dementia.

The correct answer is: F-MRI

Question **26**

Not answered

Marked out of 1.00

Flag question

Which imaging technique uses non-ionising radiation to look at regional blood flow changes based on oxygen metabolism?

Select one:

- MRI scan
- PET Scan
- SPECT scan
- fMRI scan
- CT Scan

Your answer is incorrect.

Functional MRI (fMRI) is based on the same technology as the MRI (a strong magnetic field and radio waves to produce highly-detailed images of tissue). The fMRI goes further by detecting high levels of brain activity by measuring the blood flow to the region. This is achieved by detecting changes in oxygen rich vs oxygen poor regions of the brain-the differing level of oxygen in the haemoglobin changes the effect the magnetic field has on the haemoglobin .

The correct answer is: fMRI scan

Question **27**

Not answered

Marked out of 1.00

Flag question

Sally's father died of Huntington's disease when Sally was 13. Now aged 33, she has decided to get tested because she has noticed that her limbs are making unusual movements, beyond her control. She tests positive for Huntington's. Five years later her movement problems have become significantly worse and are affecting her functioning. Which medication is she most likely to be prescribed to help with her Huntington's related movement disorder?

Select one:

- Levodopa
- Risperidone
- Citalopram
- Diazepam
- Tetrabenazine

Your answer is incorrect.

Tetrabenazine is an approved drug for this condition. It acts as a dopaminergic antagonist, but also has effects on other monoamines, leading to their reversible depletion. It may also be used for other hyperkinetic disorders such as Tourette's syndrome and Tardive Dyskinesia.

The correct answer is: Tetrabenazine

Question **28**

Not answered

Marked out of 1.00

Flag question

Mr Brown has had alcohol dependence for the past three decades. His daughter brought him to the A&E in an agitated state. The A&E staff have arranged for a set of bloods to be taken, including liver function tests. You are informed that his daughter gave him 10mg of Diazepam (her own prescription) to help calm her father down before he arrived in A&E. As part of your onward management plan, you wish to take into consideration the protein binding and metabolic properties of Diazepam. Which one of the following statements is true:

Select one:

- Diazepam has protein binding of 65% and no active metabolites
- Diazepam has protein binding of 98% with temazepam as the main active metabolite
- Diazepam has protein binding of 98% with the main active metabolite being desmethyldiazepam
- Diazepam has protein binding of 65% with the main active metabolite being desmethyldiazepam
- Diazepam has protein binding of 98% with no active metabolites

Your answer is incorrect.

Diazepam has protein binding of 98% with the main active metabolite being desmethyldiazepam. Minor active metabolites are Oxazepam and Temazepam

The correct answer is: Diazepam has protein binding of 98% with the main active metabolite being desmethyldiazepam

Question **29**

Not answered

Marked out of 1.00

Flag

The pseudocholinesterase deficiency is associated with increased sensitivity to which of the following chemicals?

Select one:

- Fluoxetine
-

question

Donepezil

- Propofol
- Sodium Valproate
- Carbamazepine

Your answer is incorrect.

It is advised that Donepezil is avoided in people with confirmed pseudocholinesterase deficiency. There are case reports of adverse effects, particularly when certain muscle relaxants (Succylcholine and Mivacurium) are used.

The correct answer is: Donepezil

Question **30**

Not answered

Marked out of 1.00

Flag question

Caffeine is a commonly used and socially acceptable psychoactive substance. Which of the following statements is incorrect regarding caffeine?

Select one:

- It can cross the placenta?
- It is metabolised similarly in all ethnic groups?
- It is metabolised by CYP1A2?
- Saliva concentration is half of plasma level
- It is more damaging to foetus in rapid metabolisers than slow.

Your answer is incorrect.

Caffeine is present in serum and saliva is similar concentrations. Caffeine is primarily metabolised by CYP1A2 and also, in smaller amounts, by CYP2E1, CYP3A4, CYP2C8, CYP2C9. Konje and Cade (BMJ - 2008) showed the association between caffeine and foetal growth restriction was stronger in women with a faster compared to a slower caffeine clearance. It crosses the placenta freely. There are ethnic differences between the speed of caffeine metabolism.

The correct answer is: Saliva concentration is half of plasma level

Question **31**

Not answered

Marked out of 1.00

Flag question

The blood brain barrier has an important role in only permitting selected molecules through it and into the cerebral and cerebellar tissue. Which of the following does it allow through?

Select one:

- Noradrenaline
- Lipophilic big un-ionised molecules
- L-DOPA
- Hydrophilic molecules
- Histamine

Your answer is incorrect.

L-DOPA can pass through the blood brain barrier via a specialized transmembrane transporter called L-type amino acid transporter. Molecules can pass through the blood brain barrier if they are small, lipophilic (un-ionised), a gas or if they have a specific transporter.

The correct answer is: L-DOPA

Question **32**

Not answered

Marked out of 1.00

Flag question

Serotonin is thought to be an important neurotransmitter in the aetiology and treatment of mental disorder. Which of the following enzymes are not involved in its metabolism?

Select one:

- COMT
- DOPA decarboxylase
- MAO
- Tryptophan hydroxylase
- Aldehyde dehydrogenase

Your answer is incorrect.

COMT (Catechol-O-methyltransferase) is one of several enzymes that degrade catecholamines (such as dopamine, adrenaline, and noradrenaline), catecholestrogens, and various drugs and substances having a catechol structure. Serotonin does not have a catechol structure (benzene with two hydroxyl side groups at carbons 1 and 2) and a side-chain amine.

The correct answer is: COMT

Question **33**

Not answered

Marked out of 1.00

Flag question

Duloxetine is an antidepressant used in the UK. Which of the following is true about it?

Select one:

- Patients with gastric bypass surgery may need a lower dose
- It is a dual action antidepressant
- Smoking has no impact on the plasma levels and efficacy
- It has a half-life of around 36 hours
- Doses of 120mg are routinely used for neuropathic pain

Your answer is incorrect.

Explanation:

- Duloxetine, along with venlafaxine, is sometimes termed as a dual action antidepressant (acts on both serotonin and noradrenaline).
- The elimination half-life of duloxetine is approximately 12 hours.
- Usual dosage ranges are 40-60mg/day in 1-2 doses for depression, 60mg once daily for

diabetic peripheral neuropathy and fibromyalgia, and 40mg twice daily for stress urinary incontinence.

- Studies have not demonstrated increased efficacy of duloxetine beyond 60mg/day. Some patients may require up to or more than 120mg/day, but clinical experience is quite limited with high dosing. In neuropathic pain and fibromyalgia, doses above 60mg/day have been associated with increased side effects without an increase in efficacy.
- Plasma levels of duloxetine can be reduced by up to 50% in smokers (Pg 857, Maudsley 14th) and 42% lower after gastric bypass surgery (Pg 823, Maudsley 14th). Higher dose may be necessary for these patients to achieve therapeutic benefits.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 265-266.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 823, 857.

The correct answer is: It is a dual action antidepressant

Question **34**

Not answered

Marked out of 1.00

Flag question

Laura is prescribed a new antidepressant, vortioxetine. Which receptor does vortioxetine exert its action on?

Select one:

- 5HT2
- D2
- 5HT6
- D4
- 5HT3

Your answer is incorrect.

Explanation:

Vortioxetine causes serotonin transporter (SERT) inhibition as well as having antagonist actions at 5HT3 and 5HT7 receptors, with agonist actions at 5HT1A receptors and weak partial agonist to antagonist actions at 5HT1B/D receptors. This unique blend of pharmacological actions leads to downstream release of many different neurotransmitters, and these actions hypothetically lead to antidepressant effects in unipolar depression, characterised by robust pro-cognitive actions, especially improving processing speed.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 315.

The correct answer is: 5HT3

Question **35**

Which one among the following is a pharmacodynamic concept?

Not answered

Marked out of 1.00

Flag question

Select one:

- Distribution
- Metabolism
- Agonism
- Absorption
- Excretion

Your answer is incorrect.

Explanation:

Pharmacokinetics is the study of how the body acts upon drugs, especially to absorb, distribute, metabolise, and excrete them. Pharmacodynamic actions at specific drug targets are known as the mechanism of action of psychotropic drugs, and account for most of the therapeutic effects and side effects of drugs. Agonism, receptor half-life, occupancy, dose-response relationship are pharmacodynamic concepts.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 49.

The correct answer is: Agonism

Question **36**

Not answered

Marked out of 1.00

Flag question

Sasha has returned home from a night out, during which she drank 12 units of alcohol. Her housemate is very worried about her, as she appears barely conscious, and decides to take her to the Emergency Department. You are aware that alcohol is eliminated in a constant amount per unit time, and that this constant does not vary according to plasma concentration of alcohol. What type of elimination kinetics is this describing?

Select one:

- Zero order
- Linear
- First order
- Second order
- Exponential decay

Your answer is incorrect.

In zero order kinetics (also known as non linear kinetics) the elimination rate saturates at high drug concentrations, as occurs with alcohol, phenytoin, and salicylates, among others.

The correct answer is: Zero order

Question **37**

Not answered

Marked out of 1.00

The effectiveness of certain psychotropic medications can be affected by pharmacokinetics. Which of the following is an aspect of pharmacokinetics?

Select one:

Flag question

Agonism
- Dose-response relationship
- Half-life
- Absorption
- Chelation

Your answer is incorrect.

Explanation:

Pharmacokinetics is the study of how the body acts upon drugs, especially to absorb, distribute, metabolise, and excrete them. Agonism, half-life, dose-response relationship and chelation are pharmacodynamic concepts.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 49.

The correct answer is: Absorption

Question **38**

Not answered

Marked out of 1.00

Flag question

Psychotropic medications will sometimes enter adipose tissue and become concentrated there. What properties of a drug aid its movement into fat tissue?

Select one:
- Hydrophobia
- Methylation
- Oxidation
- Lipophilic
- Chemical polarity

Your answer is incorrect.

Lipophilic compounds enter adipose tissue more easily. The other states described would make crossing a cell membrane more difficult.

The correct answer is: Lipophilic

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